# CHECKLIST OF BBM104 SPRING 2025 PROJECT ASSIGNMENT 4

Please try the following test cases and drop your notes under each bullet point in **BLUE** color. You can see some examples below:

- Check that functionality x works.
  This functionality works like a charm.
- Check that functionality y works.
  This functionality has some flaws, the system crashes when it is tried to do operation namely t.
- Check that functionality z works.
  This functionality does not work at all.

You can add extra test cases please insert them as in **RED** color, your notes are still in **BLUE**. This paper is also for noting which parts of your system does not work well, so you can create hypothetic test cases that make your system not respond as it has to and explain the problem.

## Test Cases

- Contents of the assets folder must be same as the one provided at Piazza.
  All assets are placed in the correct folder. Images for walls, tanks, bullets, and explosions are correctly loaded from assets/.

- Open the terminal at your main folder (the folder that your code needs to be compiled and run)
  I opened the terminal and changed directory to src because Main.java is located there. From there, I compiled and ran the project successfully.

- Run the following commands:
  - javac -version (must be resulted same as “javac 1.8.0_xxx”)
    javac 1.8.0_431

  - java -version (must be resulted same as “java version "1.8.0_xxx"”)
    java version "1.8.0_441"

- Compile your code and then run it.
  Code compiles without any errors. Game runs smoothly when launched.

- Show that the game opens without any crash. Your system must also read the assets by itself.
  Game window opens successfully with no crashes. Assets (walls, tanks, bullets) load automatically at startup.

- Show that Player movement and shooting works correctly. The player tank should be correctly controlled with the movement buttons and it should fire correctly with the “x” button. Firing should not look erratic(no shooting rapid fire with no

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wait time). Tank movement should be accompanied by a simple animation of the tank using the images provided.
Player moves with arrow keys. Shooting works using the 'X' key. A cooldown is implemented to limit shooting rate. The tank shows a basic animation when moving.

- Show that Enemy movement and shooting works correctly. Enemies should move randomly or semi randomly and they should shoot at semi random intervals. Keep in mind that tanks should not move too erratically(constantly changing directions multiple times in a single second) and they should not be able to fire again right after firing. Enemies should spawn correctly.
  Enemies spawn and move in semi-random directions. They fire bullets at random intervals. Their behavior is not erratic, and they do not overlap with player or walls.

- Show that bullets behave correctly. Bullets should hit and destroy the correct type of tanks depending on who fires them. Bullet hit should be accompanied by appropriate explosion image.
  Bullets disappear when hitting walls or tanks. Player bullets destroy enemies and increase score. Explosions are displayed upon hit. Enemy bullets can hit the player and reduce lives.

- Show that walls are implemented. Walls should prevent tanks from passing through and cause bullets to disappear. There should be walls surrounding the map and inside the area itself.(You can place them as you wish)
  Walls are placed both around the map and within it. Tanks and bullets cannot pass through them. Bullet explosions appear on collision.

- Scoring works correctly and lives decrease as necessary.
  Each destroyed enemy increases score by 10. Player lives decrease when hit by enemy bullets. Score and lives are updated on the screen.

- Pause menu works correctly. Game can be resumed or closed or restarted
  Pressing 'P' pauses/unpauses the game. “PAUSED” label appears. Game logic freezes while paused.

- Game Over triggers correctly and Game Over Menu works correctly. Game can be closed or restarted.
  When player lives reach 0, "GAME OVER" label and restart instructions are displayed. Pressing 'R' restarts the game. Pressing 'ESC' exits.

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- Moreover, please narrate the critical parts of your code (design), please do not narrate all of it, just narrate it for 1-2 minutes as it is just for understanding the code by your own perspective.
  **Code Design Narration**

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In this project, I structured my code around the `Main` class which handles the window, game logic, and rendering. I used a 2D array to define the wall layout, and I placed the player and enemy tanks on the map accordingly.

I implemented movement for both the player and enemies. The player is controlled with arrow keys, and shooting is done with the "X" key. Enemy tanks choose random directions and shoot at semi-random intervals.

I used an `AnimationTimer` for the game loop. It updates movement, bullet positions, and collision checks every frame. When bullets hit tanks or walls, I show explosion effects and update score or lives depending on who was hit.

I also implemented pause and game over screens with proper key bindings. The camera follows the player tank by shifting the layout of the root group.

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## INCOMPLETE PARTS

- There is no spawn protection for the player. If the enemy fires a bullet towards the respawn location before the player dies, the player can get hit immediately after respawning.
- During the 1-second respawn delay after the player dies, it is still possible to move the camera as if the player is alive. The tank is invisible, but camera follows key presses, which may be confusing.
- Enemy tanks are guaranteed not to spawn inside walls, but there is no check for overlapping with other tanks during respawn. As a result, newly spawned tanks can occasionally appear very close to each other or the player.
- Bullet-to-bullet collision is **not implemented**. Bullets do not detect or respond to collisions with other bullets.( Was not required in the specs, but could be considered a future improvement.)

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## RESTRICTIONS

- Your demo **must contain sound**, you can use Zoom's video recorder for this purpose, be sure about that you are sharing your **whole screen, not just the application.**
- Your demo video **cannot be longer than five minutes, four minutes for the GUI and one minute for code,** (3-4 minutes for the game and 1-2 minutes for the code is also OK but the total length of your video cannot be longer than five minutes) it cannot exceed five minutes even a little second, and your video must be in real speed, please do not speed up or down your videos.
- Everything about your **demo video must be in English** (you do not have to set your system language to English for sure). Your video must be fluent. It is not assumed that you are a native speaker, but it is known that your English level is at least B1+ (according to prep class), so, it is requested from you to speak at least as a B1+ speaker, little mistakes about your speech are going to be discarded but avoid mistakes as much as you can.
- **Do not do these operations too fast or too slow,** make them at the speed that a normal user **(who is not very familiar with your system and tries to read texts)** does.
- **Show the things that make your system crash or things that you couldn't done in demo properly** but show them after the requirements of the demo finishes, you can skip some of the requirements by saying that you did not implement it, so that your demo video becomes more fluid, less erroneous, but note that you must show your erroneous parts at the end.
- You must both narrate your GUI and code.
- **It is your own responsibility to show all the functionality of your system,** so if you have not mentioned about a functionality that is a request of this project or part of your extra feature, your code may be graded as low as zero for that part as there is no working example. So, please do not feel limited with given conditions, show your all work to get full credit, that means, if you have not shown a concept at your demo video, you will not have any right to say that **"But it worked on my machine!",** because this demo video is a chance for you to show that your code works on your machine well, but note that you can still have right to objection for the parts that are not mentioned at PDF, Piazza Notes, Q&As; say that there is a bug at a functionality and it is neither mentioned at PDF, Piazza, Q&A, nor mentioned during your implementation and you did not notice that it does not work on other machines, then for sure you can object for it by clipping a video that shows it works on your machine.

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